Acute Care of Esophageal Cancer

Acute Presentation of Esophageal Cancers

- Bleeding

- Minimal Dysphagia
- Obstruction (unable to maintain hydration)
- Perforation

Bleeding Esophageal Cancer

Triage questions:

- Has EGD been done? Pathology results?
- Presence of metastatic disease on CT chest/abdomen/pelvis?
- Upper GI bleeding requiring transfusion? Is bleeding persistent?
- Dysphagia
- Can patient maintain hydration orally?

Bleeding Esophageal Cancer

Bleeding is almost always in the context of anticoagulation

Some (without dysphagia) are T2 N0 $\Rightarrow$ primary surgery.

Most are T3 $\Rightarrow$ preoperative chemo $\pm$ radiation.

Endoscopic Ultrasound (EUS) is necessary to distinguish T2 from T3

- Available only at CMC Main
- Usually can be done as an outpatient

500+ Esophagectomies at CMC since 2007 and none performed emergently for bleeding

Esophageal Cancer with Minimal Dysphagia (without bleeding)

- Outpatient PET
- Outpatient EUS

Hospitalization (or transfer to CMC) not needed

Endoscopic ultrasound (EUS)

Distinguishes localized tumors (T2 N0) who may be candidates for primary surgery from patients with locally-advanced cancers T3 who need preoperative chemotherapy $\pm$ radiation.

Patients with T2 tumors generally have minimal dysphagia and frequently present with bleeding during anticoagulation or surveillance for Barrett's esophagus.

EUS has limited utility in the staging of patients with esophageal cancer who present with dysphagia (Mansfield et al. 2017) (Findlay et al. 2015) (Radlinski et al. 2020) (Ripley et al. 2016) and is not necessary.

Patients with dysphagia who can maintain oral hydration

Clinically staged as T3 and fit into one of two caregories:

- Patients with metastatic disease (liver, lung, or extra-regional lymphadenopathy with cervical or para-aortic nodes). Treatment is chemotherapy with radiation for palliation of dysphagia or treatment of symptomatic bone metastasis

- Patients without metastatic disease are staged *Locally Advanced* and treated with preoperative chemotherapy ± radiation followed by surgery.

Surgery as primary therapy is not indicated except for cancer perforation.

Esophageal Obstruction with Negative Biopsy

Endoscopy with biopsy is foundational for patients with esophageal obstruction.

Majority will have evidence of malignancy on EGD. Negative endoscopic biopsies in the patient with an obstructing esophageal stricture are unusual but not rare.

Options for re-biopsy include:

- Neonatal (5mm) endoscope
- Dilation followed by biopsy
- Endoscopic ultrasound with FNA.

These options are available at CMC Main and may not be available at regional facilities.

(Faigel et al. 1998)

Axial Imaging

CT chest/abdomen/pelvis with IV contrast for initial staging

PET/CT Scan

- Needed for complete staging
- Rarely needed in the acute setting
- Required for radiation therapy planning

PET/CT Prep:

- NPO for 4 hours (except water)
- No IV dextrose for 4 hours prior
- Blood sugars controlled

Esophageal Obstruction (can't maintain hydration)

Patients with esophageal obstruction who can't hydrate need inpatient feeding tube

Enteral Feeding Tube

Metastatic disease: PEG

- Most patients can be treated with a neonatal scope + 20Fr Pull PEG

No metastatic disease

- Lap gastrostomy preferred
- Lap jejunostomy never the wrong answer

Laparoscopic Gastrostomy for Resectable Esophageal Cancer

Gastrostomy

- Can injury right gastroepiploic
- Can damage future conduit

Jejunostomy

- Safer option (no damage to stomach)
- Less convenient

⇒ Laparoscopic gastrostomy can be advantageous if placed properly

Conduit Construction

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Laparoscopic Gastrostomy

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Laparoscopic Gastrostomy Techniques

Neonatal (5mm) endoscope can be helpful

After laparoscopic ports placed, drop insufflation pressure to 4mmHg

Place stomach away from the right gastroepiploic artery in the "Northern Hemisphere" on a saggital plane

More proximal placement is better *as long as* tube is at least 3cm from costal margin

Operative Description of Laparoscopic Gastrostomy

Laparoscopic Jejunostomy

Traditional method of nutritional support for patients with esophageal obstruction.

Avoids the risk of injury to the future gastric conduit at the time of esophagectomy.

Operative Descripton of Laparoscoic Jejunostomy

Postoperative Care for Jejunostomy

Endoscopic stents

Appropriate with metastatic disease (Stage IV) and those who will never be candidates for surgery.

In patients who are potentially candidates for esophagectomy, stents are generally avoided to prevent complications of surgery through fibrosis of the esophagus proximal to the tumor.

Stent vs Gastrostomy Feeding Tube

A retrospective study of patients with esophageal cancer and dysphagia (Min et al. 2017) found that treatment with gastrostomy was associated with longer survival than those treated with a endoscopic stent and had higher albumin levels and less need for re-intervention.

(Min et al. 2017)

Percutaneous Endoscopic Gastrostomy

Indicated for patients with Stage IV disease

- Allows bolus feeding (more convenient than a jejunostomy)

Avoided is patients with potentially resectable diseaase

- Injury to right gastroepiploic artery

- Seeding of gastrostomy tract with cancer

Central Venous Port for Chemotherapy

Can facilitate chemotherapy administration

Not essential in acute management

Younger patients can receive chemoradiation with peripherap IVs

Can usually be deferred to outpatient care once seen by treating medical oncologist

Perforated Esophageal Cancer

Most common contexts:

- Endoscopy
- Endoscopic ultrasound
- Dilation of esophagus

Endoscopic stenting may be difficult due to underlying stricture

(Abu-Daff et al. 2016)

Diagnosis

CT esophagram with water soluble contrast is the diagnostic study of choice. This should include CT neck and/or CT abdomen/pelvis depending upon the suspected level of perforation.

(Suarez-Poveda et al. 2014)

Pneumomediastinuim $\neq$ Esophageal perforation

65 patients transferred with suspected esophageal perforation found that 24 patients were ultimately found not to have a perforation.

Among patients for whom pneumomediastinum was their sole criteria for transfer, only 57% were ultimately found to have a perforation.

Among patients with a pleural effusion prior to transfer, 83% were ultimately found to have a perforation.

Patients with a lactate greater than 1.6 were 11 times more likely to have a perforation.

(Madsen et al. 2023)

Perforated Esophageal Cancer - Triage Questions

Triage questions:

- Has EGD been done? Pathology results?
- Spontaneous or s/p procedure?
- Has patient had prior radiation therapy?
- CT Chest/abdomen/pelvis
- Pleural effusion or pneumothorax?
- Pneumoperitoneum or peritoneal fluid?

Acute Management

- IV hydration
- Correction of electrolyte abnormalities
- IV antibiotics
- CT Esophagram with water soluble contrast
- CT abdomen/pelvis
- CT neck (if suspicion of cervical perforation)

Inpatient Management

- Emergent drainage of pleural effusions
- Emergent laparoscopic placement of drains for evidence of intra-abdominal perforation

Cervical esophageal perforation

Contained perforations of the cervical esophagus can be treated with NPO and IV antibiotics.

Uncontained perforations need surgical exploration. In this case, nutritional support with gastrostomy is indicated.

Thoracic esophageal perforations

Pittsburgh esophageal perforation scoring system: (Abbas et al. 2009) Validation of Pittsburgh scoring system (Schweigert et al. 2016)

A series from Birmingham UK of 74 esophageal perforations over 15 years include 13 patients with perforated cancers, of whom 7 were treated with endoscopic stenting (**charalampakis1710?**). These 13 patients had a 1-year mortality of 77%

Conservative management of Perforated Esophageal Cancer

There is little experience in the literature as clinical reports exclude patients with perforated cancers.

A report from Kenya of 10 perforations among 492 dilations of esophageal cancer, of whom 9 were successfully treated with endoluminal stents (White et al. 2003). Stenting of perforated esophageal cancer is limited to case reports (Kobayashi et al. 2019)

Emergent Esophagectomy

Emergency esophagectomy is the traditional approach to perforated esophageal cancer given the difficulty in stenting an area of tumor (or stricture). Short-term goals are fluid resuscitation, IV antibiotics, and drainage of the pleurae (and abdomen).

Emergent esophagogastrectomy via an abdominal and thoracic approach is the conventional approach in a fit patient. For frail patients, an transabdominal approach with esophagogastrostomy is less morbid (Gillen et al. 2009) although placing the patient at higher risk of postoperative reflux.

Orientation Manual

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